# Biology and Environmental Sciences Comprehensive Program Review
# AY 2020-2021

Faculty: Alison Varty, Dave Clarke, Jenny Heath, and Ann Klein

## 1 – Description of Program

***Organization***

Staffing: The COS Science Department is currently comprised of four full-time faculty members, three biology instructors (Varty, Clarke, and Klein) and one chemistry instructor (Heath). Two of our full-time faculty members also oversee environmental sciences (Varty and Heath). Our program is in the process of hiring a full-time physics and astronomy instructor for spring 2021. It is important to mention that with the spring 2019 retirement of our full-time physics and engineering instructor, the program no longer offers engineering courses. The program also employs four biology and environmental sciences part-time instructors teaching both general education and courses for science majors, including geography. We have one part-time geology instructor and one part-time physics instructor. In addition to teaching personnel, the science program is supported by one ten-month laboratory technician.

Structure: All members of the program report to the Vice President of Academic Affairs, who is also functioning as our dean.

***Whom we serve***

The Science Department serves three groups of students:

- STEM majors – one of the fastest growing areas nationwide
- General Education (GE) – non-science majors are exposed to the process of science and fact-based inquiry, an especially important task in our society today
- Pre-health students – all health-related programs require a set of science classes as prerequisite for program admission

| ASTR | 93 |
|---|---|
| Unknown / Not Collected | 20 |
| MT. SHASTA HIGH | 14 |
| Golden Eagle Charter | 6 |
| Outside Now | 4 |
| YREKA HIGH | 4 |
| Banks High School | 3 |
| WEED HIGH | 3 |
| BUTTE VALLEY HIGH | 2 |
| CORNING HIGH | 2 |
| PLEASANT VALLEY HIGH | 2 |
| (Other (<2) | 33 |

| CHEM | 160 |
|---|---|
| Unknown / Not Collected | 36 |
| YREKA HIGH | 30 |
| MT. SHASTA HIGH | 21 |
| WEED HIGH | 8 |
| Golden Eagle Charter | 7 |
| DUNSMUIR HIGH | 5 |

| BIO | 1367 |
|---|---|
| Unknown / Not Collected | 256 |
| YREKA HIGH | 156 |
| MT. SHASTA HIGH | 116 |
| Outside Now | 69 |
| WEED HIGH | 44 |
| Banks High School | 38 |
| Golden Eagle Charter | 33 |
| ETNA UNION HIGH SCHOOL | 28 |
| SHASTA HIGH | 24 |
| DUNSMUIR HIGH | 20 |
| FOOTHILL HIGH | 18 |
| ANDERSON HIGH | 16 |
| Foreign Country | 16 |
| WEST VALLEY HIGH | 14 |
| Squalicum High School | 13 |
| BUTTE VALLEY HIGH | 12 |
| CENTRAL VALLEY HIGH | 11 |
| ENTERPRISE HIGH | 11 |
| MATTOLE VALLEY CHARTER (#159) | 11 |

| | |
|---|---|
| ETNA UNION HIGH SCHOOL | 5 |
| BUTTE VALLEY HIGH | 4 |
| DISCOVERY HIGH | 2 |
| Outside Now | 2 |
| Other (<2 students) | 40 |

| **ENGR** | **4** |
|---|---|
| Unknown / Not Collected | 1 |
| HAPPY CAMP HIGH | 1 |
| LEILEHUA HIGH SCHOOL | 1 |
| BIG VALLEY HIGH | 1 |

| **PHYS** | **363** |
|---|---|
| Unknown / Not Collected | 60 |
| YREKA HIGH | 46 |
| MT. SHASTA HIGH | 39 |
| WEED HIGH | 19 |
| Banks High School | 16 |
| ETNA UNION HIGH SCHOOL | 10 |
| Golden Eagle Charter | 10 |
| Outside Now | 9 |
| DUNSMUIR HIGH | 6 |
| ANDERSON HIGH | 4 |
| CENTRAL VALLEY HIGH | 4 |
| Foreign Country | 4 |
| Squalicum High School | 4 |
| BUTTE VALLEY HIGH | 3 |
| CORNING HIGH | 3 |
| HAPPY CAMP HIGH | 3 |
| LAGRANGE HIGH SCHOOL | 3 |
| Troup County Comprehensive Hs | 3 |
| TULELAKE HIGH | 3 |
| Other (<3 students) | 114 |

| | |
|---|---|
| RED BLUFF HIGH | 11 |
| MODOC HIGH | 10 |
| Other (<10 students) | 440 |

| **ENVS** | **174** |
|---|---|
| Unknown / Not Collected | 34 |
| MT. SHASTA HIGH | 26 |
| YREKA HIGH | 18 |
| WEED HIGH | 10 |
| Golden Eagle Charter | 8 |
| Outside Now | 6 |
| Banks High School | 5 |
| JEFFERSON HIGH (CONTINUATION) | 3 |
| BISHOP KELLY HS | 2 |
| DUNSMUIR HIGH | 2 |
| ETNA UNION HIGH SCHOOL | 2 |
| FALL RIVER JUNIOR-SENIOR HIGH | 2 |
| HAPPY CAMP HIGH | 2 |
| MCCLOUD HIGH | 2 |
| MERRILL F. WEST HIGH | 2 |
| Squalicum High School | 2 |
| Other (<2 students) | 48 |

| **GEOL** | **246** |
|---|---|
| Unknown / Not Collected | 97 |
| YREKA HIGH | 25 |
| MT. SHASTA HIGH | 15 |
| Golden Eagle Charter | 10 |
| WEED HIGH | 9 |
| Outside Now | 7 |
| Banks High School | 6 |
| DUNSMUIR HIGH | 3 |
| ETNA UNION HIGH SCHOOL | 3 |
| FOOTHILL HIGH | 3 |
| Foreign Country | 2 |
| MATTOLE VALLEY CHARTER (#159) | 2 |
| Stissing Mountain Jr-Sr Hs | 2 |
| Zane Trace High School | 2 |
| Other (<2 students) | 60 |

| | **ASTR** | **BIO** | **CHEM** | **ENGR** | **ENVS** | **GEOL** | **PHYS** |
|---|---|---|---|---|---|---|---|
| F | 40 43.01% | 889 65.0% | 101 63.13% | 1 25.00% | 83 47.70% | 138 56.10% | 169 46.56% |
| M | 53 56.99% | 464 33.9% | 59 36.88% | 3 75.00% | 89 51.15% | 103 41.87% | 189 52.07% |
| N | | 12 0.9% | | | 2 1.15% | 5 2.03% | 5 1.38% |
| Unknown/Not Collected | | 2 0.1% | | | | | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Total | 93 | 1367 | 160 | 4 | 174 | 246 | 363 |

| | ASTR | | BIO | | CHEM | | ENGR | | ENVS | | GEOL | | PHYS | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| African American | 4 | 4.30% | 37 | 2.71% | 2 | 1.08% | 1 | 25.00% | 2 | 1.15% | 5 | 2.03% | 25 | 6.89% |
| American Indian/Alaskan Native | 2 | 2.15% | 41 | 3.00% | 6 | 3.23% | | | 7 | 4.02% | 8 | 3.25% | 9 | 2.48% |
| Asian | 1 | 1.08% | 28 | 2.05% | 5 | 2.69% | | | 3 | 1.72% | 3 | 1.22% | 5 | 1.38% |
| Filipino | 1 | 1.08% | 9 | 0.66% | 13 | 6.99% | | | | 0.00% | 1 | 0.41% | 1 | 0.28% |
| Hispanic | 9 | 9.68% | 73 | 5.34% | 26 | 13.98% | | | 9 | 5.17% | 8 | 3.25% | 22 | 6.06% |
| Pacific Islander | | | 12 | 0.88% | | | | | 2 | 1.15% | 1 | 0.41% | 3 | 0.83% |
| Two or More Races | 21 | 22.58% | 240 | 17.56% | 21 | 11.29% | 1 | 25.00% | 33 | 18.97% | 28 | 11.38% | 67 | 18.46% |
| Unknown/Not Reported | 5 | 5.38% | 52 | 3.80% | 8 | 4.30% | | | 5 | 2.87% | 17 | 6.91% | 15 | 4.13% |
| White | 50 | 53.76% | 875 | 64.01% | 105 | 56.45% | 2 | 50.00% | 113 | 64.94% | 175 | 71.14% | 216 | 59.50% |
| Total | 93 | | 1367 | | 186 | | 4 | | 174 | | 246 | | 363 | |

The tables above show the number of students served by each discipline within our department and where are students are coming from. Biology is serving the most students (56.8%) followed by Physics (15.0%) and then Geology (10.2%). We primarily serve students from our local high schools, Mount Shasta, Weed, and Yreka High Schools.

The tables above also describe the demographics of our students. Our students are predominately white (63.1%), followed by students with two or more races (16.9%), and then Hispanic students (6.0%). Most of our students identify as females (62.7%) while 39.9% of our students identify as males and a small portion do not identify as males or females.

***Primary purpose***

Our general purpose is to:

- Prepare STEM majors for transfer and careers in science fields,
- Increase science literacy with our GE offerings, and
- Prepare pre-health students for health-related degrees and licensing programs.

***The kinds of services we provide:***

The Science Department offers courses in seven disciplines including astronomy, biology, chemistry, engineering, environmental sciences, geology, geography and physics. Offerings within these disciplines include general education courses which allow students to get to know the discipline while completing science general education requirements. Degree specific courses are also offered in support of the degree programs housed within the Science Department including Associate in Science degrees in Engineering, Environmental Science, and Physical Science and Associate Degrees for Transfer in Geology, Geography, Physics and Liberal Arts and Sciences – Natural Sciences. The Science Department also offers courses for pre-health students which are required prerequisites for admission to health-related programs.

***How we provide services:***
The tables and analyses below quantify when our courses are offered during the week and day, and the amount that are credit degree applicable verses credit non-degree applicable courses.

***Breakdown of the classes offered in your areas within the Program Review cycle.***

| Term | Day | Evening | Internet | Weekend | Grand Total |
|---|---|---|---|---|---|
| Fall 2016 | 17 | 2 | 3 | 2 | 24 |
| Spring 2017 | 17 | 2 | 6 | 1 | 26 |
| Summer 2017 | 0 | 0 | 3 | 0 | 3 |
| Fall 2017 | 16 | 5 | 5 | 0 | 26 |
| Winter 2018 | 0 | 0 | 3 | 0 | 3 |
| Spring 2018 | 14 | 3 | 7 | 0 | 24 |
| Summer 2018 | 1 | 0 | 3 | 0 | 4 |
| Fall 2018 | 19 | 2 | 4 | 0 | 25 |
| Winter 2019 | 0 | 0 | 4 | 0 | 4 |
| Spring 2019 | 15 | 0 | 5 | 0 | 20 |
| Summer 2019 | 0 | 0 | 2 | 0 | 2 |
| Fall 2019 | 12 | 5 | 4 | 0 | 21 |
| Winter 2020 | 0 | 0 | 1 | 0 | 1 |
| Spring 2020 | 17 | 2 | 5 | 0 | 24 |
| Grand Total | 128 | 21 | 55 | 3 | 207 |
| Percent | 61.84% | 10.15% | 26.57% | 1.45% | |

The majority of our courses offered between Fall 2016 and Spring 2020 were offered during the day (61.8%), 10% are offered at night. Slightly more than one-quarter of our courses offered were internet courses while 1.4% were weekend courses. However, if you count our botany courses as weekend courses that number jumps to nearly 4%.

| | Credit - Degree Applicable | Credit - Not Degree Applicable | Grand Total |
|---|---|---|---|
| ASTR | 4 | 0 | 4 |
| BIO | 66 | 14 | 80 |
| CHEM | 14 | 0 | 14 |
| ENGR | 1 | 0 | 1 |
| ENVS | 6 | 0 | 6 |
| GEOL | 18 | 10 | 28 |
| PHYS | 30 | 0 | 30 |
| Grand Total | 139 | 24 | 163 |

All of the courses we offered between fall of 2016 and spring 2020 were credit courses and the majority of those courses, 85%, were degree applicable. The only non-degree applicable courses offered in the sciences are Biology Prep and the weekend field courses in Geology.

***Describe how your curriculum is up-to-date and needs based***

All of our classes target one or more of the groups of students that we earlier identified as serving. Science general education is a core of every GE program in the world. All of our 'majors' classes address the needs of students interested in obtaining an AS degree in STEM or transferring. Other courses including

Microbiology, Anatomy, and Physiology serve students in our pre-health program. These three classes, especially, are a link between the Liberal Arts and the Career and Technical programs in a way that other LAS class are not.

Our scheduling has been responsive to those needs, in that we have increased and maintained our offerings of microbiology, anatomy and physiology as the need has risen. We make every effort to ensure that STEM students can count on a consistent, dependable schedule of the majors classes. And we constantly mix and match our GE offerings to ensure a variety of classes offered at different times and in different modalities.

We constantly review all of our classes for currency and make curricular adjustments as they present themselves. We also are not 'locked in' to existing structures. For example, in response to the loss of our geology instructor and his mostly GE-serving classes, we are exploring having ENVS 1002 count as a physical science for GE purposes. And we are decommissioning one core biology class to replace it with another that better fits our students' needs. We are also planning to begin offering a non-major, non-laboratory chemistry course.

Our responsiveness to the needs of our students is also reflected in the absence of ADTs in the two majors that would have the highest volume – chemistry and biology. The high unit load of these ADTs, coupled with our small size precluding us from offering every required class, every semester places an unreasonable burden on those students. So instead, we direct those students to pursue the AS in Natural Sciences. They're still able to get all the classes they need, but at their own pace and working around our schedule.

We currently offer ADTs in geology, geography, and physics. Although we no longer offer an ADT in engineering, we still offer the full range of transfer courses, including higher level math, that support these students' transfer goals. Many of the these required ADT courses are supported by part-time faculty members.

## 2 – External Factors

a) Budgetary constraints or opportunities
   - Our budget places constraints on what we can achieve, but we feel our current department budgets are adequate. However, as we anticipate greater competition from other schools for online students moving forward, we think it would be wise for the institution to devote more of the instruction budget to hiring part-time and full-time instructors to teach online science courses, particularly for non-majors. This may be an opportunity moving forward.
   - Our department is deficient in computers. Physics had an entire classroom set that has become too old to update. Requests for replacements for those and for a shared set of 24 computers for the building have not been entirely filled (we did receive 9 computers last academic year). Our need for a set of 32 computers to share will be greater once we are back in the classroom. Since moving more content online, we have found more simulations and activities that require computers in the classroom and still feel it is essential for our students to be able to input, graph and analyze data during class.

b) Competition from other institutions
   - We anticipate that because science faculty across the country developed online courses during the COVID-19 pandemic, there will be more science courses being taught online. Thus, we see our competition in this arena increasing.

c) Requirements of four-year institutions
   - From our perspective, these do not seem to be changing. Ease of transferability of online courses has likely improved. There are a handful of disciplines in our area that require higher level math

courses to transfer. The Science program would like to express the importance of supporting its chemistry, physics and engineering students, by committing to offer higher level math courses that will fit into the science students' scheduling needs, by working with the math program. It will ensure that although we do not offer ADTs in some of these areas, our students will be successful in completing four-year institution transfer goals.

d) Requirements of prospective employers
   - Many pre-health programs have exercised flexibility and are allowing students to complete some prerequisite lab courses online during the pandemic. Our understanding is that these exceptions are temporary and that face-to-face laboratory classes will be required to satisfy these prerequisites once it is safe to hold these classes.

## 3 – Progress on Outcomes Assessment

a) Please summarize the progress your unit has made on program and/or course level SLO measures you have applied since your last Program Review
   - We have been tracking and analyzing student attainment of learning outcomes informally. It informs large and small changes we make to our courses regularly. However, we don't have clarity on the current institutional process for tracking these assessments.

b) Please describe any program/course and/or instructional improvements made by your unit as a result of the outcomes assessment process.
   - We have made enormous changes to our courses recently as a result of SLO assessment and the COVID-19 pandemic. These changes include:
     - Clarifying rubrics and assignment guidelines
     - Changing the amount of time devoted to certain subjects
     - Greatly modifying assignments to meet learning objectives in the online or remote environment

c) Please describe the program's assessment plan for the next four years.

We need to engage in a meaningful and formalized process and will do so once that process has been delineated.

## 4 – Institutional Program Effectiveness Indicators

To demonstrate our effectiveness as a program, we present our retention, success, FT/PT ratio, and our FTES data from fall 2016 to spring 2020 below.

**Retention**

| | Retention Count | Retention % | Total Count |
|---|---|---|---|
| **Fall 2016** | **351** | **82.98%** | **423** |
| BIO | 167 | 76.96% | 217 |
| CHEM | 34 | 80.95% | 42 |
| ENVS | 30 | 81.08% | 37 |
| GEOL | 82 | 93.18% | 88 |
| PHYS | 38 | 97.44% | 39 |
| **Spring 2017** | 334 | **78.59%** | 425 |
| BIO | 201 | 75.85% | 265 |
| CHEM | 19 | 90.48% | 21 |
| GEOL | 45 | 75.00% | 60 |
| PHYS | 69 | 87.34% | 79 |
| **Fall 2017** | **333** | **80.63%** | **413** |
| BIO | 191 | 76.10% | 251 |
| CHEM | 22 | 88.00% | 25 |
| ENVS | 16 | 69.57% | 23 |

| | | | |
|---|---|---|---|
| GEOL | 44 | 89.80% | 49 |
| PHYS | 60 | 92.31% | 65 |
| **Spring 2018** | **339** | **84.54%** | **401** |
| BIO | 195 | 79.27% | 246 |
| CHEM | 43 | 93.48% | 46 |
| GEOL | 48 | 87.27% | 55 |
| PHYS | 53 | 98.15% | 54 |
| **Fall 2018** | **280** | **84.08%** | **333** |
| BIO | 151 | 80.75% | 187 |
| CHEM | 28 | 87.50% | 32 |
| ENGR | 4 | 100.00% | 4 |
| ENVS | 21 | 100.00% | 21 |
| GEOL | 38 | 79.17% | 48 |
| PHYS | 38 | 92.68% | 41 |
| **Spring 2019** | **274** | **79.88%** | **343** |
| BIO | 179 | 74.58% | 240 |
| CHEM | 4 | 100.00% | 4 |
| ENVS | 21 | 95.45% | 22 |
| PHYS | 70 | 90.91% | 77 |
| **Fall 2019** | **288** | **80.00%** | **360** |
| BIO | 206 | 78.93% | 261 |
| CHEM | 14 | 100.00% | 14 |
| ENVS | 32 | 72.73% | 44 |
| GEOL | 25 | 92.59% | 27 |
| PHYS | 11 | 78.57% | 14 |
| **Spring 2020** | **266** | **67.68%** | **393** |
| BIO | 189 | 66.55% | 284 |
| CHEM | 21 | 61.76% | 34 |
| ENVS | 22 | 73.33% | 30 |
| GEOL | 6 | 60.00% | 10 |
| PHYS | 28 | 80.00% | 35 |

Across the sciences, retention has averaged around 80% or greater since the Fall 2016 semester. COVID-19 negatively impacted our retention in Spring 2020 when retention dropped to 67.7%. With the decline in retention, there was also a decline in success as shown in the table below. Our goal for the next 4 years is to see our retention rates rise back to the previous norm of 80%.

**Success**

| | Success Count | Success % | Total Count |
|---|---|---|---|
| **Fall 2016** | **284** | **67.14%** | **423** |
| BIO | 134 | 61.75% | 217 |
| CHEM | 26 | 61.90% | 42 |
| ENVS | 21 | 56.76% | 37 |
| GEOL | 70 | 79.55% | 88 |
| PHYS | 33 | 84.62% | 39 |
| **Spring 2017** | **292** | **68.71%** | **425** |

| | | | |
|---|---|---|---|
| BIO | 180 | 67.92% | 265 |
| CHEM | 14 | 66.67% | 21 |
| GEOL | 31 | 51.67% | 60 |
| PHYS | 67 | 84.81% | 79 |
| **Fall 2017** | **275** | **66.59%** | **413** |
| BIO | 148 | 58.96% | 251 |
| CHEM | 21 | 84.00% | 25 |
| ENVS | 11 | 47.83% | 23 |
| GEOL | 39 | 79.59% | 49 |
| PHYS | 56 | 86.15% | 65 |
| **Spring 2018** | **294** | **73.32%** | **401** |
| BIO | 169 | 68.70% | 246 |
| CHEM | 33 | 71.74% | 46 |
| GEOL | 40 | 72.73% | 55 |
| PHYS | 52 | 96.30% | 54 |
| **Fall 2018** | **235** | **70.57%** | **333** |
| BIO | 121 | 64.71% | 187 |
| CHEM | 25 | 78.13% | 32 |
| ENGR | 4 | 100.00% | 4 |
| ENVS | 16 | 76.19% | 21 |
| GEOL | 31 | 64.58% | 48 |
| PHYS | 38 | 92.68% | 41 |
| **Spring 2019** | **246** | **71.72%** | **343** |
| BIO | 160 | 66.67% | 240 |
| CHEM | 4 | 100.00% | 4 |
| ENVS | 15 | 68.18% | 22 |
| PHYS | 67 | 87.01% | 77 |
| **Fall 2019** | **246** | **68.33%** | **360** |
| BIO | 171 | 65.52% | 261 |
| CHEM | 14 | 100.00% | 14 |
| ENVS | 28 | 63.64% | 44 |
| GEOL | 22 | 81.48% | 27 |
| PHYS | 11 | 78.57% | 14 |
| **Spring 2020** | **238** | **60.56%** | **393** |
| BIO | 168 | 59.15% | 284 |
| CHEM | 19 | 55.88% | 34 |
| ENVS | 20 | 66.67% | 30 |
| GEOL | 4 | 40.00% | 10 |
| PHYS | 27 | 77.14% | 35 |

Our average success rate from fall of 2016 to spring 2020 was 68.4%. From the data, we see that students are typically slightly more successful in the spring compared to fall semester. We see this pattern in some of our sequenced courses; after students make it through the first course, they tend to do well in the second. The exception to the pattern above was spring 2020 when success rates were unusually low, 60.56%. We attribute this to the disruptions to everyday life and instruction associated with the COVID-19 pandemic.

Our goal for the next 4 years would be to return to a more typical success rate for our department. When we exclude spring 2020 from the average success rate, the average is 69.5%. We feel this is a reasonable future goal.

**PT/FT ratio**

We have seen some significant shifts in our FT/PT faculty ratio from Fall 2016. Whereas part-time instructors taught 25% of our courses, all within the discipline Biology, in Fall 2016, they taught closer to 50% of our courses in Fall 2019. At that point, less than half of the courses taught by part-time faculty were in Biology.

We attribute these shifts to several factors. First, we lost two full-time faculty members in our department within this time period: one in Geology and one in Physics (who also taught Engineering courses). We also gained a full-time faculty member in Biology and lost several part-time faculty members in Biology.

The data indicate that, instructor availability has been, to a large extent, dictating which courses we offer. Ideally, student demand would drive our offerings.

| **Fall 2016** | Part-time | Full-time | Totals | **Fall 2019** | Part-time | Full-time | Totals |
|---|---|---|---|---|---|---|---|
| Totals | 6 | 18 | 24 | Totals | 11 | 10 | 21 |
| ASTR | 0 | 2 | 2 | ASTR | 1 | 0 | 1 |
| BIO | 6 | 4 | 10 | BIO | 4 | 8 | 12 |
| CHEM | 0 | 3 | 3 | CHEM | 0 | 1 | 1 |
| ENVS | 0 | 2 | 2 | ENVS | 1 | 1 | 2 |
| GEOL | 0 | 4 | 4 | GEOL | 3 | 0 | 3 |
| PHYS | 0 | 3 | 3 | PHYS | 2 | 0 | 2 |

The loss of two full-time faculty members in Physical Sciences has also impacted our FTES. The table below shows that FTES in Physics dropped by more than half of historic norms in AY 2019-2020. The same thing occurred in AY 2018-2019 with the loss of our full time Geology instructor. It appears that some of these FTES may be being picked up in Biology and Environmental Science. We know from our colleagues in counseling that students are currently struggling to take Physical Science GE courses at COS and suspect that we will see an increase in FTES in physics with the addition of a full-time Physics and Astronomy instructor in spring 2021.

| | **1st Year** | **2nd Year** | **3rd Year** | **4th Year** | **Grand Total** |
|---|---|---|---|---|---|
| **Row Labels** | | | | | |
| ASTR | 1.1 | 0 | 6.09 | 2.56 | 9.75 |
| BIO | 84.63 | 98.34 | 88.93 | 114.01 | 385.91 |
| CHEM | 16.79 | 20.61 | 9.72 | 13.8 | 60.92 |
| ENGR | 0 | 0 | 0.43 | 0 | 0.43 |
| ENVS | 7.4 | 4.77 | 9.05 | 15.8 | 37.02 |
| GEOL | 10.9 | 9.45 | 4.68 | 4.04 | 29.07 |
| PHYS | 22.04 | 27.32 | 24.27 | 9.27 | 82.9 |
| **Grand Total** | **142.86** | **160.49** | **143.17** | **159.48** | **606** |

## 5 – Other Unit-Specific Quantitative and Qualitative Research

**Degrees awarded**

| | 2017 | 2018 | 2019 | 2020 | Grand Total |
|---|---|---|---|---|---|
| LAS Natural Science | 22 | 27 | 43 | 28 | 120 |
| Biology, General | 1 | 3 | 1 | 1 | 6 |
| Chemistry, General | 1 | 2 | 0 | 0 | 3 |
| Environmental Science | 0 | 0 | 1 | 1 | 2 |
| Geology | 0 | 0 | 1 | 0 | 1 |
| Physics, General | 0 | 1 | 4 | 0 | 5 |

The most common degree we are granting in the sciences is the LAS Natural Science degree. In 2017, 2018, 2019, and 2020, we awarded 120 of these degrees out of 137 total degrees awarded over the same time period. The other degrees we grant, Biology and Chemistry (both of which are no longer offered), Environmental Science, Geology, Geography and Physics collectively totaled 12.4% of the science degrees awarded.

The AS-Natural Science degree is popular with students. This degree is flexible, students can choose a curriculum within the degree that prepares them for numerous transfer programs and/or careers, which likely explains the popularity. The concept of a flexible degree such as this one is nicely aligned with Guided Pathways.

## 6 – Evaluation

a) Representativeness of population served- The science faculty (and the faculty as a whole) are less diverse than the student body. We feel greater racial and ethnic diversity would bolster our goals of serving minority students well and we are committed to working with appropriate partners to support activities to recruit a diverse faculty.

b) Alternative modes and schedules of delivery (e.g., online, hybrid, early morning, evening services)
We have increased the number of courses that we offer online and hybrid. However, we feel that these courses will be even more popular and available after the pandemic. We think it would be wise to thoughtfully increase our online and hybrid offerings. This will require recruiting more part-time and full-time science instructors.

c) Partnerships (internal and external)
We have a strong relationship with the US Forest Service for Work Experience students. We have been working with our new board member Debbie Derby to broaden this relationship to include other local agencies and natural resources organizations. This is going well. However, we need more institutional support for Work Experience to keep in going. Instructors used to get paid twice as much for the same amount of work, we also don't get paid according the contract timelines or rates, and it's a lot of work. Even though we know this has generated amazing opportunities for our students, we are considering dropping the Work Experience requirement from the Environmental Science degree if we do not get more support and/or compensation for these efforts.

d) Innovation and implementation of best practices

We continue to innovate and strive to implement best practices in our courses. Some examples of this include:

- Revamped the Microbiology course
- Integrated the atomic flame absorption instrument into chemistry courses
- Re-vamped Environmental Science discussions to promote critical thinking and the use of sound information when stating opinions.
- Made greater use of digital resources such as micrographs, simulations, and online, authentic data sets.

Technology limitations have presented challenges to implementing some of the best practices in science education:

- Students need access to computers during class to conduct simulations, analyze data, generate tables and figures, etc.
- Student need access to computers with functioning microphones and web cameras limit our ability to assess and engage students remotely.

Although the science department has taught online and hybrid courses for years and we feel strongly that these opportunities are important for our students, the recent switch to remote instruction has highlighted the fact that the remote modality is not appropriate for some of our courses. It hinders the implementation of best practices by limiting the quality of laboratory activities students can participate in and limiting their group interactions.

e) Efficiency in operations

We have been efficient in sharing a single laboratory technician as a department. However, our technician's efficiency has been compromised by the following issues:

- Our laboratory technician often does not receive timely budget information from the Instruction Office
- Our laboratory technician does not have a credit card and this means we experience delays in ordering and receiving supplies
- The science faculty play a limited role in evaluating the laboratory technician's performance

f) Efficiency in resource use

We could be more efficient in our use of instructional supplies including consumables, if we had more access to digital technology in the classroom.

g) Staffing

We struggle with staffing and as the FTES and PT/FT ratio data have suggested, our offerings are being driven by available instructors instead of student need for courses. We are lacking in the Physics and Geology. This has severely limited our ability to offer General Education courses in Physical Sciences and to offer many of the courses in Physics and Geology for science majors. As always, our ability to offer our full suite of classes depends on our ability to recruit and retain part-time faculty – an especially difficult task in our county. In point of fact, we do not have enough part-time instructors in biology, resulting in two of the three FT biology faculty teaching overload that they really don't want. It would not be a stretch to contend that we could use a fourth biology FT as well.

With that point made, Clarke will be retiring sometime soon, possibly as early as spring 2022. We are concerned that the college will seek to return to a biology faculty size of two people and force the reassignment of the remaining faculty into classes they are not best suited for, removing their expertise from the classes they *are* suited for, and reducing the number of sections of microbiology, anatomy and

physiology, which always fill. So, even though it's not a 'goal' for this AY, and given the delay in replacing the full-time Physics Instructor, we would like to get it out there – COS *must* replace Clarke when he retires

At one point, the college was exploring the creation of an Earth Science full time faculty position. Based on the great need our students have for Physical Science GE courses and the fact that Geology and Environmental Science have served 17% of our students (we were not given data on Geography, but we know our Geography offerings have been healthy as well), we suggest that this notion be revisited.

A handful of our other (outside of biology) program disciplines rely heavily on part-time instructors. We currently have one part-time geology instructor. When we lose this instructor, we will no longer be able to offer the geology ADT. We are in a similar situation with geography courses.

h) Participation in shared governance (e.g., do unit members feel they participate effectively in planning and decision-making?)

Although we feel that the departmental budget development process has been inclusive, we have little say in budgeting decisions outside of our departmental budgets. Furthermore, from our experience serving on many shared governance committees, we feel that faculty participation on shared governance committees is not valued and that many important planning and budget decisions are made without adequate input.

i) Professional development and training

The equity and diversity training we have received this academic year has been excellent. Other than that, it remains difficult to obtain other high-quality professional development experiences given the institution's budget for faculty professional development.

## 7 – Vision

We envision ourselves:

- Reaching more students through welcoming online and face-to-face courses and departmental environment (more online courses and equity and inclusion improvements)
- Using more technology to effectively engage students in our online and face-to-face courses. This vision will be supported by the addition of science building laptops, used in all courses.
- Meeting the course offering needs of our majors and general education students (increased staffing and planning).

## 8 – Progress on Prior Goals

This is the first 4-year action plan our unit has completed. Thus, we have not previously made 4-year goals. We have included the goals from the most recent Biology and Environmental Science Program review. The text in italics is our status update.

- <u>Recruiting and retaining part-time faculty</u>. We have lost many quality and 'maxed out' part-time faculty to other opportunities and retirement over the past four years (Bill Odegard, Eveline Robinson, Ariel Halpern, Logan Smith, Joe Champagne) and have been unable to find replacements. Although the physics program has been functioning with one part-time instructor (that has varied over the past years with 3 different instructors) the program is not providing a full range of courses for majors and engineering students.
  *This situation has not improved since we wrote this goal was drafted. If we want to meet students needs for Physical Science General Education and majors courses (particularly in the Physical Sciences), this needs to continued work.*

- No department credit card for our technician. Having a credit card would streamline our purchases and require less work by the business and instructional offices. She is extremely busy; this may free up some of her time to allow her to provide our students with more support. *This concern carries over from last year, it has not been addressed and continues to create inefficiencies within our department.*
- Delayed budgetary information. We are not notified of our budget until well after the semester starts, making prioritization and obtaining supplies for the first labs of the semester problematic. *This concern carries over from last year, it has not been addressed.*
- Timing of instructional equipment requests. Non-instructional areas are allowed to request funds at any time of year, while we are limited to CQIPS submitted with program review. *This concern carries over from last year, it has not been addressed.*
- Computer access. Increasingly, many laboratory activities require computer access. The ASC computers are seldom available. *We asked for a full set of 24 to 'equip' one full lab class; we received 9. The need for laptops has increased. Thus, we have included a CQIP for 32 more laptop computers.*
- Professional development. Funds for professional development are inadequate. *This concern carries over from last year, it has not been addressed.*
- Work Experience. We mentioned above that Biology Work Experience is required for Environmental Science students. We feel this experience and the related connections to local environmental scientists is crucial for our graduates. However, the turnover in the Work Experience Coordinator position makes helping students obtain these experiences difficult. We highly recommend that the college invest significantly in this program. *This is still an issue and it (in addition to instructor compensation issues mentioned elsewhere) are prompting us to consider eliminating the Work Experience component of the Environmental Science degree*.

## 9 – Four-Year Action Plan

This analysis of departmental data and the status of previous goals has led us to propose the following action plan:

Goal 1: Provide our face-to-face students with access to cutting-edge in-class computer simulations, models, and programs that will allow them to input, analyze, and graph data they are collecting in class.

- Objective: Obtain a set of laptop computers to be shared by the faculty for use in the Science Building and devise a system for managing their use building-wide

Goal 2: More deeply incorporate equity and inclusion into our department.

- Objectives: We have been inspired by our recent professional development experiences to strive to ensure that we are creating inclusive environments in our building and classrooms. The outcome will look different for all of us, but could include:
    - A building display that highlights the work of women and minorities in science
    - Inclusion and diversity statements for our syllabi
    - The incorporation of new content into our courses that highlights the struggles and/or contributions of women and minorities in science
    - Supporting efforts to recruit diverse science faculty members

Goal 3: Increase access to our courses for members of our community and beyond.

- Objective: We will use the skills and knowledge we have gained teaching remote and online courses during the COVID-19 pandemic to offer more hybrid and online science courses.

- This will require meeting a series of shorter-term outcomes:
    - Evaluate which of our courses lend themselves well to the online format and make the appropriate updates to the course outlines

Support our Administration in their efforts to:

- Locate and retain part-time instructors to teach some of the selected courses online
- Ensure the currency of online training for all online instructors in our department
- Ensure all faculty teaching online for our department have the equipment they need to effectively teach online

Goal 4: Ensure we are meeting our students' needs for courses in the Sciences

- Deficiencies have already been identified below in Geology and in the Physical Sciences GE area. Our biggest issue here is staffing.
- Objectives:
    - Work with our Dean to ensure an early planning process to replace retiring full-time faculty members
    - Actively recruit and retain part-time faculty members
    - Complete the hiring process of a full-time Physics faculty member
    - Further examine the need for an additional Geology/Earth Science full-time faculty member

## Signature Page

We the undersign certify that the insert program name here program review process included broad dialogue about student learning, program effectiveness and resource allocation.

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Principal Preparer

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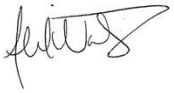
Contributor



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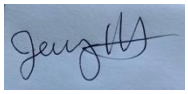

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